

Study Guide

this is a place to start, it is not ALL inclusive

learn the big picture first; then dive into the details; seek understanding not memorization

L18: Immune 1

1. Define immunity
2. What does the immune system protect us from?
3. Describe the function of the various types of white blood cells.
4. What are the external barriers to infection?
5. What is an antigen? Give examples.
6. Distinguish between primary and secondary lymphoid organs? What is lymphatic tissue and where is it found?
7. List all cells of the immune system and describe their primary functions.
8. List the chemical mediators of the immune system and describe their primary functions.
9. What is the difference between the specific and nonspecific immune systems?
10. Name the components of the nonspecific immune system and how they defend us from invading pathogens.
11. **Short Answer:** Describe the process of inflammation and the 4 signs of inflammation. Explain why inflammation causes the 4 signs?

L18: Immune 2

12. What are the different mechanisms used by the immune system "kill" invading cells?
13. How do bacteria, viruses and cancer cells make us sick?
14. Differentiate inflammation, infection, injury. ***Here is the answer:*** Injury (e.g. sprained ankle) and infection both demonstrate inflammation. Inflammation during Infection is due to pathogen invasion and tissue damage; inflammation during injury is due to tissue damage only.
15. **Potential essay:** Describe the process of B & T cell activation in response to invading viruses and bacteria. How are they different and how are they the same? (HINT: you should be looking at differences and similarities in dual activation.)
16. How does body...
 - a. make millions of unique antibodies?
 - b. distinguish self from non-self?
 - c. remember pathogens?
17. Differentiate antigen, antibody, antibiotic.
18. Relate immunization, allergies, & autoimmune diseases to what you know about the immune system.
19. Why is colostrum so important to a newborn?
20. Explain how vaccinations work.
21. **Short answer:** Explain why a person becomes **SICK** after the first exposure to a pathogen and then becomes **IMMUNE** and will not get sick the second time they see the same pathogen.

22. Distinguish between passive natural immunity, passive artificial immunity, active natural immunity and active artificial immunity. Give an example of each.

This information will be gone over in the "Blood Lab"

23. **Short answer:** Explain why must patients be blood typed before receiving donor blood?
24. List the blood types, their antigens, their antibodies and what donor blood types they may receive.
25. Define the following: universal donor and universal recipient in blood donations, agglutination, transfusion reaction, hemolytic disease of the newborn.